

Program 3: Containerize a Dedicated Python Flask / Node.js Application

Project Structure

program-3/

|

├── requirements.txt

├── Dockerfile

└── app.py

Step 1: Create the Dockerfile

Filename: Dockerfile

Use official Python image

FROM python:3

Set the working directory

WORKDIR /app

Copy requirements file and install dependencies

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

Copy application code

COPY . .

Expose the port Flask will run on

EXPOSE 5000

Run the Flask application

CMD ["python", "app.py"]

```
1RV24MC029_CHINMAIAKAGOLLI@CHINMAI: ~/DevOps/prg3
GNU nano 7.2 Dockerfile
FROM python:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 8000
CMD ["python", "app.py"]

[ Read 8 lines ]
^H Help      ^O Write Out  ^W Where Is   ^X Cut        ^T Execute    ^C Location   ^U Undo       ^M-A Set Mark   ^_ To Bracket ^O Previous   ^B Back
^X Exit      ^R Read File  ^A Replace    ^U Paste      ^J Justify    ^_ Go To Line ^E Redo       ^M-G Copy     ^G Where Was ^M-W Next     ^F Forward
```

Step 2: Create the Flask Application

Filename: app.py

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello from Simple Flask Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
1RV24MC029_CHINMAVAKAGOLLI@CHINMAY: ~/DevOps/prg3
GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Simple Flask Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=8000)
```

[Read 11 lines]

⌘ Help ⌘ Write Out ⌘ Where Is ⌘ Cut ⌘ Execute ⌘ Location ⌘ Undo ⌘ Set Mark ⌘ To Bracket ⌘ Previous ⌘ Back
⌘ Exit ⌘ Read File ⌘ Replace ⌘ Paste ⌘ Justify ⌘ Go To Line ⌘ Redo ⌘ Copy ⌘ Where Was ⌘ Next ⌘ Forward

Step 3: Create the Requirements File

Filename: requirements.txt

Flask==2.3.3

```
1RV24MC029_CHINMAVAKAGOLLI@CHINMAY: ~/DevOps/prg3
GNU nano 7.2 requirements.txt
Flask==2.3.3
```

[Read 1 line]

⌘ Help ⌘ Write Out ⌘ Where Is ⌘ Cut ⌘ Execute ⌘ Location ⌘ Undo ⌘ Set Mark ⌘ To Bracket ⌘ Previous ⌘ Back
⌘ Exit ⌘ Read File ⌘ Replace ⌘ Paste ⌘ Justify ⌘ Go To Line ⌘ Redo ⌘ Copy ⌘ Where Was ⌘ Next ⌘ Forward

Step 4: Build the Docker Image

Run the following command in your terminal (inside the project directory):

`docker build -t program-3 .`

Step 5: Run the Docker Container

Map port **5000** inside the container to **5000** on the host:

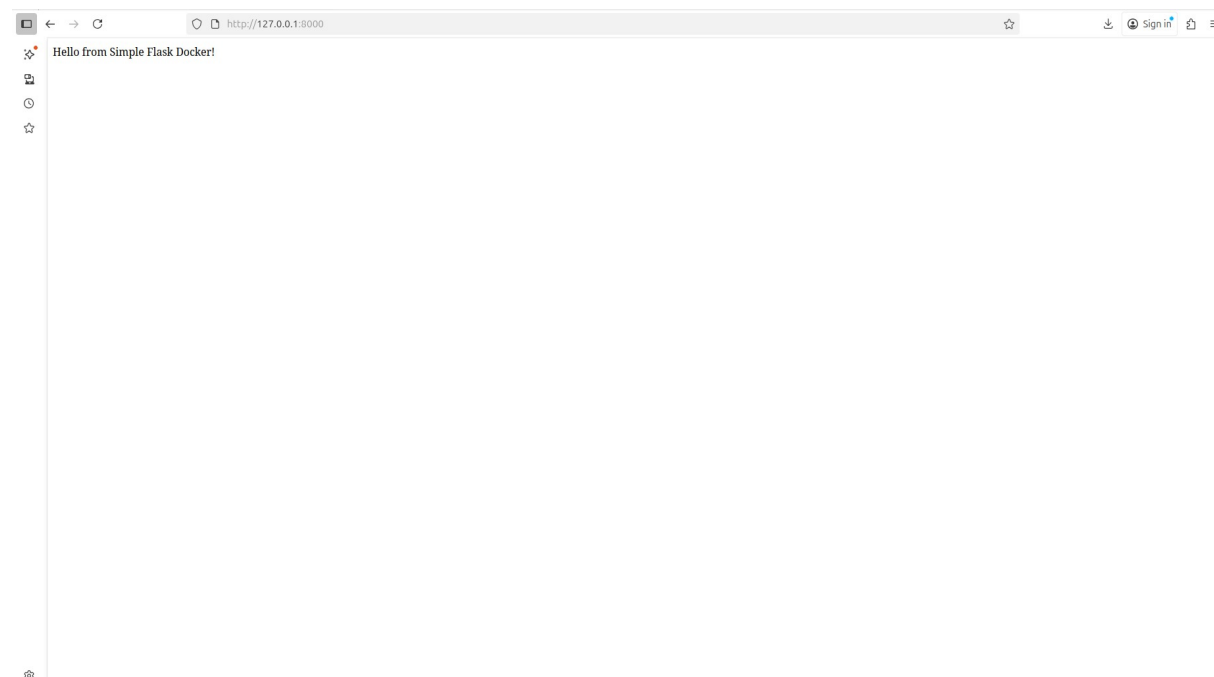
`docker run -p 5000:5000 program-3`

```
TRV24MC029_CHINMAYAKAGOLLI@CHINMAY: ~/DevOps/prg3
Step 2/7 : WORKDIR /app
--> Using cache
--> d885d4bdd93f
Step 3/7 : COPY requirements.txt .
--> Using cache
--> 025bbc787e7e
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> d316db070f25
Step 5/7 : COPY . .
--> 2b9e03f3ad84
Step 6/7 : EXPOSE 8000
--> Running in cdba1f312b6a
--> Removed intermediate container cdba1f312b6a
--> 459fb0f1cb84
Step 7/7 : CMD ["python", "app.py"]
--> Running in 65cbea7ca602
--> Removed intermediate container 65cbea7ca602
--> 6385ecf94382
Successfully built 6385ecf94382
Successfully tagged prg3:latest
TRV24MC029_CHINMAYAKAGOLLI@CHINMAY:~/DevOps/prg3$ docker run -p 8000:8000 prg3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8000
* Running on http://172.17.0.3:8000
Press CTRL+C to quit
172.17.0.1 - - [07/Nov/2025 03:54:12] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 03:54:12] "GET /favicon.ico HTTP/1.1" 404 -
172.17.0.1 - - [07/Nov/2025 04:00:24] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 04:00:24] "GET /favicon.ico HTTP/1.1" 404 -
^C1RV24MC029_CHINMAYAKAGOLLI@CHINMAY:~/DevOps/prg3$ docker run -p 8000:8000 prg3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8000
* Running on http://172.17.0.3:8000
Press CTRL+C to quit
172.17.0.1 - - [07/Nov/2025 04:04:58] "GET / HTTP/1.1" 200 -
```

Step 6: Test the Application

Open your web browser and go to:

<http://localhost:5000/> Hello from Simple Flask Docker!



Expected Output - Hello from Simple Flask Docker!

